1. Explain qualitatively what effect the following transformations will have on the Discrete Fourier Transform of an image:

(i) Random rearrangement of the intensity values of the image pixels. **[4]**

ANSWER
This process will significantly increase the high frequency content due to considerable fluctuation between neighboring intensity values, and lower frequency components (except for DC value) will reduce in magnitude.

(ii) Convolution of the image with a Gaussian kernel of minimal (almost zero) standard deviation. **[4]**

ANSWER
This process will have nearly no effect on the DFT as the image is being convolved with just a delta function.

(iii) Addition of zero mean Gaussian noise with some standard deviation σ to the image. **[4]**

ANSWER
This process will increase the high frequency content of the image significantly, especially since the image is weaker in the higher frequencies.

(b) Consider the population of vectors $\underline{f}$ of the form

$$\underline{f}(x,y) = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \\ f_3(x,y) \end{bmatrix}.$$

Each component $f_i(x,y),\ i = 1,2,3$ represents an image of size $M \times M$ where M is even. The population arises from the formation of the vectors $\underline{f}$ across the entire collection of pixels (x,y). The three images are defined as follows:

$$f_1(x,y) = \begin{cases} r_1 & 1 \le x \le \dfrac{M}{2}, 1 \le y \le M \\ r_2 & \dfrac{M}{2} < x \le M, 1 \le y \le M \end{cases}$$

$$f_2(x,y) = \begin{cases} r_3 & 1 \le y \le \dfrac{M}{2}, 1 \le x \le M \\ r_4 & \dfrac{M}{2} < y \le M, 1 \le x \le M \end{cases}$$

$$f_3(x,y) = r_5,\ 1 \le x \le M,\ 1 \le y \le M$$

Consider now a population of random vectors of the form

$$\underline{g}(x,y) = \begin{bmatrix} g_1(x,y) \\ g_2(x,y) \\ g_3(x,y) \end{bmatrix}$$

where the vectors $\underline{g}$ are the Karhunen-Loeve (KL) transforms of the vectors $\underline{f}$.

(i) Find the images $g_1(x,y)$, $g_2(x,y)$ and $g_3(x,y)$ using the Karhunen-Loeve (KL) transform. **[6]**

Answers

ANSWER

$$f_1(x,y) = \begin{cases} r_1 & 1 \le x \le \frac{M}{2}, 1 \le y \le M \\ r_2 & \frac{M}{2} < x \le M, 1 \le y \le M \end{cases}$$

Mean value of $f_1(x,y)$ is $m_1 = \frac{r_1}{2} + \frac{r_2}{2}$. Zero-mean version of $f_1(x,y)$ is

$$f_1(x,y) - m_1 = \begin{cases} \frac{r_1}{2} - \frac{r_2}{2} & 1 \le x \le \frac{M}{2}, 1 \le y \le M \\ \frac{r_2}{2} - \frac{r_1}{2} & \frac{M}{2} < x \le M, 1 \le y \le M \end{cases}$$

$$f_2(x,y) = \begin{cases} r_3 & 1 \le y \le \frac{M}{2}, 1 \le x \le M \\ r_4 & \frac{M}{2} < y \le M, 1 \le x \le M \end{cases}$$

Mean value of $f_2(x,y)$ is $m_2 = \frac{r_3}{2} + \frac{r_4}{2}$. Zero-mean version of $f_2(x,y)$ is

$$f_2(x,y) - m_2 = \begin{cases} \frac{r_3}{2} - \frac{r_4}{2} & 1 \le y \le \frac{M}{2}, 1 \le x \le M \\ \frac{r_4}{2} - \frac{r_3}{2} & \frac{M}{2} < y \le M, 1 \le x \le M \end{cases}$$

Mean value of $f_3(x,y)$ is r_5. Zero-mean version of $f_3(x,y)$ is $f_3(x,y) - m_3 = 0$.

Image $f_1(x,y) - m_1$ is of the form below:

A	$-A$

Image $f_2(x,y) - m_2$ is of the form below:

C
$-C$

Therefore, the image $(f_1(x,y) - m_1)(f_2(x,y) - m_2)$ is of the form below:

AC	$-AC$
$-AC$	AC

Therefore, $E\{(f_1(x,y) - m_1)(f_2(x,y) - m_2)\} = 0$, or, in other words, the covariance between $f_1(x,y) - m_1$ and $f_2(x,y) - m_2$ is 0.
Variance of $f_1(x,y) - m_1$ is $\frac{1}{2}\frac{1}{4}(r_1 - r_2)^2 + \frac{1}{2}\frac{1}{4}(r_1 - r_2)^2 = \frac{1}{4}(r_1 - r_2)^2$.
Variance of $f_2(x,y) - m_2$ is $\frac{1}{4}(r_3 - r_4)^2$.
Variance of $f_3(x,y) - m_3$ is 0.

Answers

Based on the above, the covariance matrix is $\begin{bmatrix} \frac{1}{4}(r_1 - r_2)^2 & 0 & 0 \\ 0 & \frac{1}{4}(r_3 - r_4)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ with eigenvalues $\frac{1}{4}(r_1 - r_2)^2$ and $\frac{1}{4}(r_3 - r_4)^2$. Therefore, by using the Karhunen Loeve transform we produce three new images, with one of them being 0 and the other being $f_1(x, y) - m_1$ and $f_2(x, y) - m_2$.

(ii) Comment on whether you could obtain the result of (b)-(i) above using intuition rather than by explicit calculation.

[2]

ANSWER

The above result is expected since two of the given images are constant and therefore they don't carry any information. This means that there is only one principal component in the given set.

Answers

2. (a) (i) Histogram equalize the 8×8 image shown in Figure 1 below, where the pixels take on values $\{0,1,\ldots,7\}$. Clearly state any assumptions you make. **[3]**

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 & 2 & 2 & 3 & 4 \\ 1 & 1 & 1 & 2 & 2 & 2 & 3 & 4 \\ 1 & 1 & 1 & 2 & 2 & 2 & 3 & 4 \\ 1 & 1 & 1 & 2 & 2 & 2 & 3 & 5 \\ 2 & 2 & 2 & 2 & 2 & 5 & 5 & 5 \\ 6 & 6 & 6 & 6 & 7 & 7 & 5 & 5 \end{bmatrix}$$

Answer

intensity	Number of pixels
0	8
1	20
2	17
3	4
4	3
5	6
6	4
7	2

Figure 1

$$p(0) = \frac{8}{64} = \frac{1}{8}$$
$$T(0) = p(0) = \frac{1}{8} \to 1$$
$$p(1) = \frac{20}{64}$$
$$T(1) = p(0) + p(1) = \frac{28}{64} = \frac{7}{16} \to 3$$
$$p(2) = \frac{17}{64}$$
$$T(2) = p(0) + p(1) + p(2) = \frac{45}{64} \to 5$$
$$p(3) = \frac{4}{64}$$
$$T(3) = p(0) + p(1) + p(2) + p(3) = \frac{49}{64} \to 5$$
$$p(4) = \frac{3}{64}$$
$$T(4) = p(0) + p(1) + p(2) + p(3) + p(4) = \frac{52}{64} \to 6$$
$$p(5) = \frac{6}{64}$$
$$T(5) = p(0) + p(1) + p(2) + p(3) + p(4) + p(5) = \frac{58}{64} \to 6$$

Answers

$$p(6) = \frac{4}{64}$$

$$T(6) = p(0) + p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = \frac{62}{64} \to 7$$

$$T(7) = 7$$

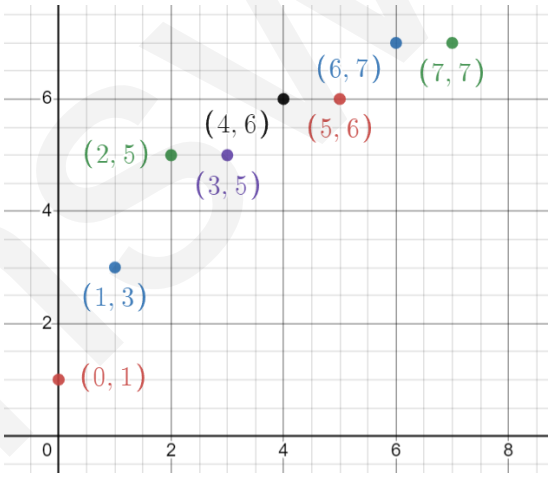
(ii) Provide the histogram of the equalised image. **[3]**

Answer

intensity	Number of pixels
0	0
1	8
2	0
3	20
4	0
5	21
6	9
7	6

(iii) Provide the transformation function that has yielded the equalized image. **[3]**

Answer



(b) Consider a digital image that has been histogram equalized. Find what happens when this image is equalized again. **[3]**

Consider an image intensity r with pdf $p_r(r) = 1$. The new intensity we get through the image equalisation transformation is

$$s = T(r) = \int_0^r p_r(w)dw = \int_0^r dw = r$$

which shows that in that case histogram equalisation has no effect.

(c) Consider the spatial filter H given by

Answers

$$H = \begin{bmatrix} -1 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

and assume a 6 −bit grayscale image I.

(i) Determine the maximum and minimum possible values that a pixel, to which this spatial filter is applied, can have. Do not apply any type of normalization. **[4]**

Answer

The image intensity takes values from 0 to 63. Therefore, the maximum value of the image after applying the filter above could possibly be $63 \times 5 = 315$ and the minimum value could possibly be $-63 \times 5 = -315$.

(ii) Propose a gray-level transformation function to ensure that any output of this filter will be a standard 6-bit grayscale image. **[4]**

Answer

The goal here is to map the minimum value of -315 to 0 and the maximum value of 315 to 63. If the original intensity is r and the new intensity is s then we can write $y = ax + b$.

$$0 = -315a + b$$
$$63 = 315a + b$$

$a = \frac{1}{10}, b = \frac{315}{10}$

Answers

3. We are given the degraded version g of an image f such that in lexicographic ordering

$$g = Hf + n$$

where H is the degradation matrix which is assumed to be block-circulant and n is the noise term which is assumed to be zero-mean, white and independent of the image f. All images involved have size $N \times N$ after extension and zero-padding.

(a) In a particular scenario the degradation process can be modelled as a linear filter with the two-dimensional impulse response given below:

$$h(x,y) = \begin{cases} 0.125 & x = 0, y = \pm 1 \\ 0.25 & x = 0, y = 0 \\ 0 & \text{elsewhere} \end{cases}$$

Estimate the frequency pairs for which Inverse Filtering cannot be applied. **[6]**

Answer

Inverse Filtering cannot be applied if $H(u,v) = 0$.

$$H(u,v) = \frac{1}{NN}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} h(x,y)e^{-j2\pi(ux/N+vy/N)} = \frac{0.125}{NN}(e^{-j2\pi u/N} + e^{j2\pi u/N} + 2)$$

$$= \frac{0.25}{NN}[\cos(2\pi u/N) + 1]$$

The values of (u,v) for which $H(u,v)$ cannot be estimated are found by: $\cos(2\pi u/N) + 1 = 0 \Rightarrow \cos(2\pi u/N) = -1 \Rightarrow u = \pi$

(b) Suppose that I am an archaeologist who had 40 ancient manuscripts similar to each other and of same size. Half of them have been permanently misplaced. The others are still in my possession. Luckily, I had acquired pictures of each of the 40 manuscripts, using a digital camera. All pictures were taken under the same lighting conditions, manuscript position, distance from the camera and camera viewpoint. Unfortunately, all pictures appeared blurred because the camera was out of focus during the acquisition. I can give you access to the exact same camera that was used for acquiring the pictures and furthermore, provide information regarding all the conditions under which the pictures were taken, so that you can reproduce the exact same conditions.

(i) Provide a method to obtain the degradation model which caused the blurring. Use the fact that many images are available, in order to remove the additive camera noise which cannot be assumed to be negligible. **[7]**

Answer

Assume that the sharp version of the digital images, which we can reacquire from the manuscripts using our digital camera, are $f_i(x,y)$, $i = 1, \ldots 20$ and the blurred version of the images (which are the ones that are available) are $g_i(x,y)$, $i = 1, \ldots, 40$. Let's denote the degradation model with $h(x,y)$.

$g_i(x,y) = h(x,y) * f_i(x,y) + n_i(x,y)$, $i = 1, \ldots 20$

We add all the noisy images to get rid of the noise.

$$\sum_{i=0}^{20} g_i(x,y) = \sum_{i=0}^{20}(h(x,y) * f_i(x,y)) + \sum_{i=0}^{20} n_i(x,y) =$$

$$\sum_{i=0}^{20} g_i(x,y) = h(x,y) * \sum_{i=0}^{20} f_i(x,y) + 0$$

$$\sum_{i=0}^{20} g_i(x,y) = h(x,y) * \sum_{i=0}^{20} f_i(x,y)$$

$$\sum_{i=0}^{20} G_i(x,y) = H(u,v) \sum_{i=0}^{20} F_i(x,y)$$

$$H(u,v) = \frac{\sum_{i=0}^{20} G_i(x,y)}{\sum_{i=0}^{20} F_i(x,y)}$$

(ii) Explain how you can use Wiener filtering to deblur the lost images, making use of the information and camera I provided and your knowledge of image processing. Assume that all available images have the same power spectrum. **[7]**

Answer

In frequency domain the Wiener filter and the restored image are:

$$W(u,v) = \frac{S_{ff}(u,v)H^*(u,v)}{S_{ff}(u,v)|H(u,v)|^2 + S_{nn}(u,v)}$$

$$\hat{F}(u,v) = \frac{S_{ff}(u,v)H^*(u,v)}{S_{ff}(u,v)|H(u,v)|^2 + S_{nn}(u,v)} G(u,v)$$

In our case $H(u,v)$ was found in (i) above. For $S_{ff}(u,v)$ we can use the power spectrum of any of the sharp images $f_i(x,y)$, $i = 1, \ldots 20$.
We apply the relationship for the restored image to all the images $g_i(x,y)$, $i = 21, \ldots, 40$.

Answers

4. (a) Consider an image whose pixel values range from 0 to 8. The probability of any pixel having a value of 0 is 0.51 and the remaining pixel values all have equal probability.

(i) Find the entropy of this image in bits/pixel. **[3]**

Answer

$H = -\sum_{i=0}^{7} p(s_i) \ln\big(p(s_i)\big) = 2.38$ bits/symbol
$p(s_0) = 0.51$
$p(s_i) = 0.07, i = 1, \dots, 7$
$s_i = i$

(ii) Construct a Huffman code for the gray levels assuming that each of the pixel values are coded separately. **[3]**

Answer

intensity	Huffman code
0	1
1	0111
2	0110
3	0101
4	0100
5	0011
6	0010
7	000

(iii) Find the average number of bits/pixel of the coded set of values. **[3]**

Answer

$l_{\text{avg}} = -\sum_{i=0}^{7} p(s_i)\,\text{length}(s_i) = 2.4$ bits/symbol

(b) For a source with q symbols, find the maximum possible value of the entropy. **[3]**

Answer

$$H = -\sum_{i=0}^{q-1} p(s_i) \ln\big(p(s_i)\big) = -\sum_{i=0}^{q-1} \frac{1}{q} \ln\left(\frac{1}{q}\right) = \sum_{i=0}^{q-1} \frac{1}{q} \ln(q) = \ln(q)$$

(c) A digital image of size 100×100 pixels has five distinct intensity levels. We aim to code this image in a lossless manner.

(i) What is the minimum number of bits required if no additional information is available? **[2]**

Answer

3 bits/pixel

(ii) What is the minimum number of bits required if the intensity levels are distributed as shown in the Table 1 below? **[3]**

Answers

Answer

intensity	Number of pixels
r_1	2000
r_2	4000
r_3	1000
r_4	?
r_5	2000

Table 1

$H = -\sum_{i=0}^{4} p(s_i) \ln\big(p(s_i)\big) = 2.12$ bits/symbol

(iii) For each of the above two cases, construct an optimal code assuming each pixel value is coded individually. **[3]**

Answer

intensity	fixed length coding
r_1	000
r_2	001
r_3	010
r_4	011
r_5	100

intensity	Huffman coding
r_1	111
r_2	0
r_3	1100
r_4	1101
r_5	10

Answers